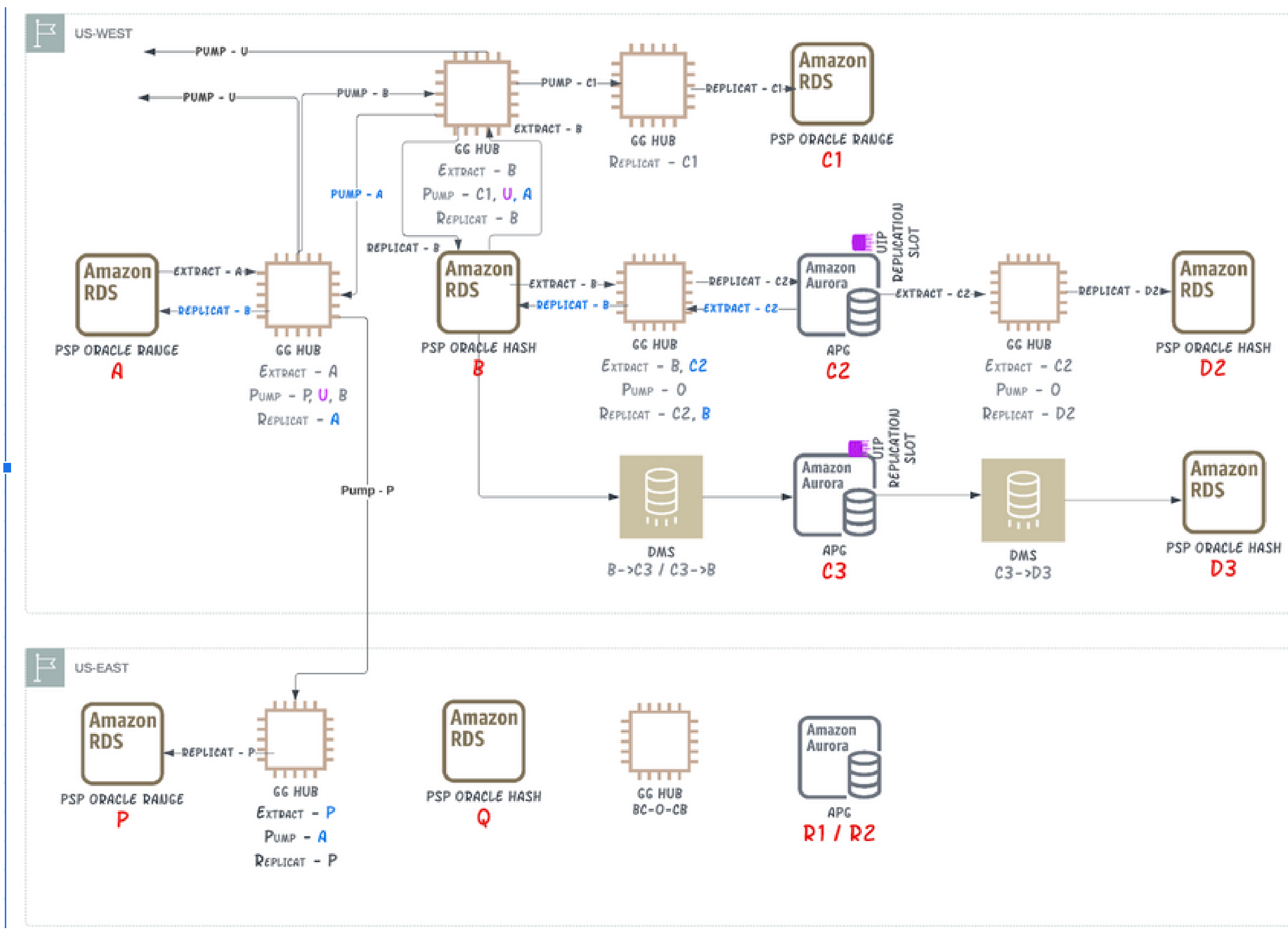
PSP Prod Monolith Aurora PostgreSQL migration

Replication Configuration



Production and DR				
West	**PSP Monolith Prod**	**Instance Name**	**East**	**PSP Monolith Prod**

A	Oracle (Range)	pspuwp01	P	Oracle (Range)
B	Oracle (HASH)	psphpp02	Q	Oracle (HASH)
B(Replica)	Oracle(HASH)	psphdg02	R1	APG (HASH) GG
C1	Oracle(Range)	psprpp01	R2	APG (HASH) DMS
C1'	Oracle(Range)	psprpp02	S1	Oracle (HASH) GG
C2	APG (HASH) GG	psp-prod-uw02,pspapg02	S2	Oracle (HASH) DMS
C3	APG (HASH) DMS	psp-prod-mon, psppp01		
D2	Oracle (HASH) GG - source C2	psphpp05		
D3	Oracle (HASH) DMS - source C3	psphpp06		
D2'	Oracle (HASH) DMS - source C2	psphpp07		

Current database object layout

Monolith

	OWNER	OBJECT_TYPE	OBJECT_COUNT
1	PSPADM	DATABASE LINK	3
2	PSPADM	FUNCTION	13
3	PSPADM	INDEX	834
4	PSPADM	INDEX PARTITION	3758
5	PSPADM	INDEX SUBPARTITION	4830
6	PSPADM	LOB	30
7	PSPADM	LOB PARTITION	464
8	PSPADM	LOB SUBPARTITION	3220
9	PSPADM	PACKAGE	3
10	PSPADM	PACKAGE BODY	3
11	PSPADM	PROCEDURE	28
12	PSPADM	SEQUENCE	76
13	PSPADM	SYNONYM	2
14	PSPADM	TABLE	383
15	PSPADM	TABLE PARTITION	1931
16	PSPADM	TABLE SUBPARTITION	805
17	PSPADM	TRIGGER	24
18	PSPADM	VIEW	5

	TABLE_NAME	No. Of Partitions
1	PSP_ENTITY_UPDATE	4
2	PSP_ENTRY_DETAIL_RECORD	187
3	PSP_FINANCIAL_TRANSACTION	94
4	PSP_FINANCIAL_TRANS_STATE	187
5	PSP_LEDGER_BALANCE	33
6	PSP_MONEY_MOVEMENT_TRANSACTION	94
7	PSP_PAYCHECK	32

8	PSP_PAYCHECK_SPLIT	32

Assessment

Summary

Monolith

Figure: Conversion statistics for database storage objects

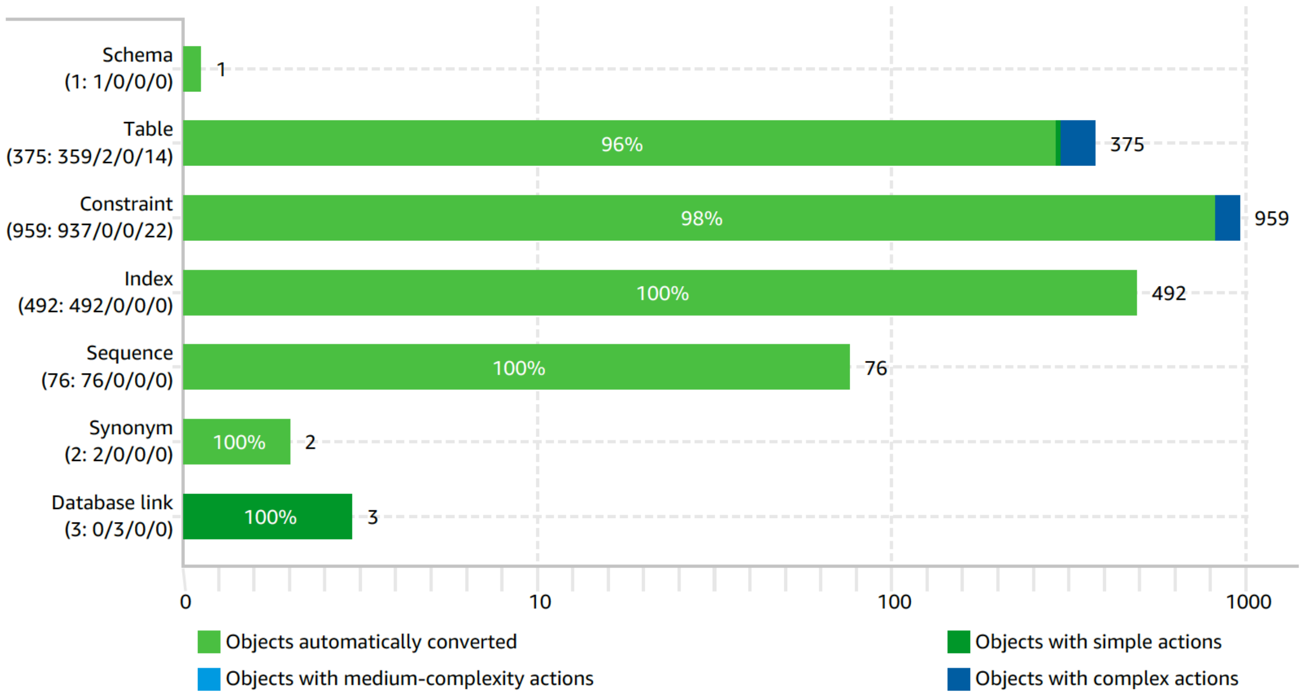
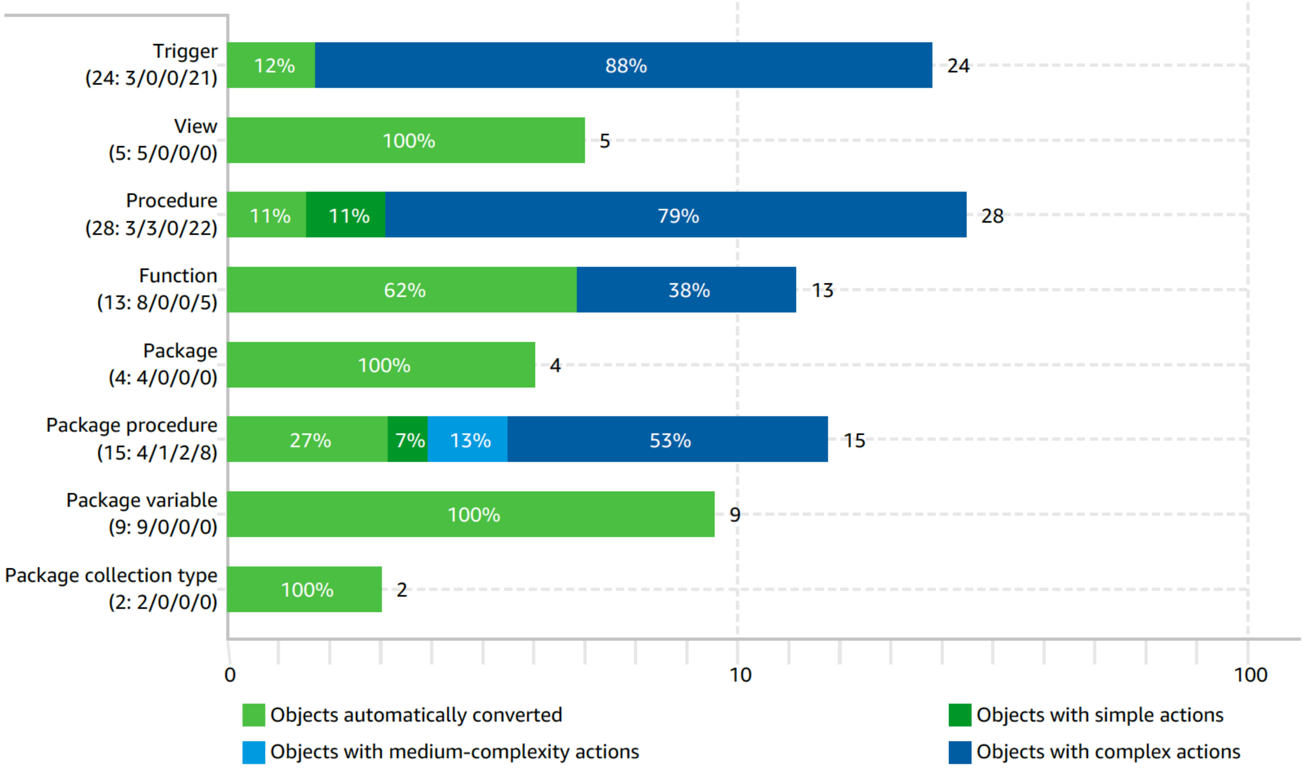


Figure: Conversion statistics for database code objects



No statistics available for database code objects for IBOB/Secondary database.

Full Report

psp_prod_monolith_sct_migration_assessment_report_aurora_postgres.pdf

Approach

- Migration would be two step approach
  - Schema refactoring on Oracle
  - Migrate to PostgreSQL
- Refactoring includes removal of **relam_id** column from PK for all tables and adding **company_fk** as PK to all partitioned tables and its childs.

Schema/Code Changes

- Primary Key
  - Remove REALM_ID and Add company_fk to PK for all partitioned tables.
  - This is to required to have hash partitioning based on company_fk.
- Partitioning
  - (Why Composite Primary Key?) This is prerequisite to partition by company_fk in Postgres. Reference
- Data Integrity
  - If Company_fk column exists then Populate company_fk column in all child tables otherwise add company_fk column in all child tables & populate it.
- Indexes
  - All Primary Key & Foreign Key indexes needs to be changed to reflect Primary Key change.
  - All other Indexes needs to be scrubbed to exclude REALM_ID.
- Performance
  - Ensure queries come to database with company_fk filter

	Database Objects	Change in Oracle Hash	Change in Postgres Hash
1	Table	• Global removal of RELAM_ID reference • PK change to composite key • Tablespace distribution of tables & Indexes	• Global removal of RELAM_ID reference • PK change to composite key
2	Indexes (Secondary & FK indexes)	• All Partitioned table PK & FK indexes changed to reflect primary key changes • Removed REALM_ID references from all indexes	• All Partitioned table PK & FK indexes changed to reflect primary key changes • Removed REALM_ID references from all indexes
3	FK Constraints	• All Partitioned table PK & FK constraints changed to reflect primary key changes • Removed REALM_ID references from all FK constraints	• All Partitioned table PK & FK constraints changed to reflect primary key changes • Removed REALM_ID references from all FK constraints
4	Tables without PK	SCHEMA_NAME,TABLE_NAME PSPADM,ARCH_PURGE_MAIN_STATS_LIST PSPADM,AS400_DROPME PSPADM,BACKUP_SCHEMA_STATS_090911 PSPADM,BACKUP_STATS PSPADM,BACKUP_STATS_032911 PSPADM,BACKUP_TAB_STATS_090911 PSPADM,CMP3$100488 PSPADM,CMP3$15077 PSPADM,CMP4$15077 PSPADM,DEL_TEST PSPADM,DISTINCT_COMPANY_FK PSPADM,EMPTY PSPADM,PAYCHECK_BACKUP_DATE PSPADM,PAYROLL_HIST PSPADM,PET_FEB27 PSPADM,PSP_COMPANY_EVENT_BKP PSPADM,PSP_COMPANY_NOTE_BKP PSPADM,PSP_EMPLOYEE_BANK_ACCOUNT_BKP PSPADM,PSP_FINANCIAL_TRANS_STATE_BKP PSPADM,PSP_LB_TMP PSPADM,PSP_LB_TMP_COMP_FK PSPADM,PSP_LB_TMP_COMP_FK_5000 PSPADM,PSP_LEDGER_BALANCE_BKP PSPADM,PSP_POSTING_RULE_MAR7 PSPADM,PSP_PROPERTY_AUDIT_BKP PSPADM,PSP_RPT_PAID_EMPLOYEES PSPADM,PSP_SAP_METHOD_CALL_BKP PSPADM,PSP_TRANSACTION_TYPE_MAR7 PSPADM,REP_TEST PSPADM,STATS_11DEC PSPADM,TEMP1_PAYCHECK PSPADM,TEMP_A_PRC_INC PSPADM,TEMP_BAD_PAYROLL PSPADM,TEMP_BKP_EDR_JUNE3RD PSPADM,TEMP_COMPANY_07152010 PSPADM,TEMP_COMPANY_DATAFILE PSPADM,TEMP_DD4V_PRC_INC PSPADM,TEMP_DD_PRC_INC PSPADM,TEMP_ENTITLE_PENDING PSPADM,TEMP_FIX_NCD PSPADM,TEMP_FIX_NCD_REACTIVATION PSPADM,TEMP_FT PSPADM,TEMP_FUTURE_PAYROLL_DATA PSPADM,TEMP_GUID_COMPANY PSPADM,TEMP_MODES_AGENCY_IDS PSPADM,TEMP_ORSTT_AGENCY_IDS PSPADM,TEMP_PAYCHECK PSPADM,TEMP_PEDR_FEB23 PSPADM,TEMP_PFTS_FEB23 PSPADM,TEMP_PFT_FEB23 PSPADM,TEMP_PMMT_FEB23 PSPADM,TEMP_PSP_PAYROLL_FRAUD_BATCH	

		PSPADM,TEMP_RAFFI_SQL_LOG PSPADM,TEMP_SUBSCRIPTIONNUMBER PSPADM,TEMP_TAX_B PSPADM,TEMP_TAX_C PSPADM,TEMP_TAX_CALC_APR12 PSPADM,TEMP_TAX_D PSPADM,TEMP_TAX_E PSPADM,TEST2 PSPADM,TEST_MMT_ATF PSPADM,TMP_AUTH_USER PSPADM,TMP_AUTH_USER1 PSPADM,TMP_PEWP_FEB27 PSPADM,TMP_PFT_FEB26 PSPADM,TMP_PMMT_FEB25 PSPADM,TMP_PSP_EDR PSPADM,TMP_PSP_MMT PSPADM,TMP_PTTMD_FEB27 PSPADM,TOAD_PLAN_SQL PSPADM,T_BACKFILL_METADATA PSPADM,T_PCEEP_OLD PSPADM,T_PDATL_OLD PSPADM,T_PPPI_OLD PSPADM,T_PPU PSPADM,T_PPU_OLD PSPADM,Z_REDEF_PAYCHECK PSPADM,Z_REDEF_PAYCHECK_SPLIT PSPADM,Z_REDEF_SRC_SYS_TRANSMISSION	
5	Tables which are not migrated/skipped	• PSP_ENTITY_UPDATE_HIST	
6			

Detailed Schema Refactor checks

- PSP Schema Refactor Checks

Manual Validations Performed

- Total object types and its count
- Invalid Objects
- Total Tables
- Partitioned Tables and its partition count
- All Primary Key Columns
- All Primary Key Indexes
- All Index types and its count
- All constraint types and its count'
  Compare All CHECK and NOT NULL COSNSTRAINTS from production
- Enabled Triggers Count
- Partitioned PK indexes and its partition count
- Local indexes and its partition count
- All FK constraint Names, FK constraint columns and respective Primary table, primary table columns
- Only FK indexes'

Schema Comparison via DB solo or liquibase

```
--PSP_FINANCIAL_TRANSACTION
create table PSPADM.PSP_FINANCIAL_TRANSACTION
(
        FINANCIAL_TRANSACTION_SEQ VARCHAR2(255) not null
                constraint SYS_C008669
                        check ("FINANCIAL_TRANSACTION_SEQ" IS NOT NULL),
        VERSION NUMBER(19) not null
                constraint SYS_C008670
                        check ("VERSION" IS NOT NULL),
        CREATOR_ID VARCHAR2(30),
```

```
        CREATED_DATE TIMESTAMP(6) not null
                constraint SYS_C008671
                        check ("CREATED_DATE" IS NOT NULL),
        MODIFIER_ID VARCHAR2(30),
        MODIFIED_DATE TIMESTAMP(6) not null
                constraint SYS_C008672
                        check ("MODIFIED_DATE" IS NOT NULL),
        REALM_ID NUMBER(19) default -1 not null
                constraint SYS_C008673
                        check ("REALM_ID" IS NOT NULL),
        FINANCIAL_TRANSACTION_AMOUNT NUMBER(19,4),
        SETTLEMENT_DATE TIMESTAMP(6),
        SETTLEMENT_TYPE_CD VARCHAR2(255)
                constraint C_PSP_FINANCIAL_TRANSACTION0                        check (SETTLEMENT_TYPE_CD IN
('EFE', 'ACH', 'Wire', 'Cash', 'CheckType', 'Other', 'ApplyForward', 'HPDE', 'TakenOnReturn', 'EFTPS',
'EFTPSDirectDebit', 'EDI')),
        CREDIT_BANK_ACCOUNT_TYPE VARCHAR2(255)
                constraint C_PSP_FINANCIAL_TRANSACTION1                        check (CREDIT_BANK_ACCOUNT_TYPE
IN('Company', 'Employee', 'Intuit', 'TaxAgency', 'Payee')),
        DEBIT_BANK_ACCOUNT_TYPE VARCHAR2(255)
                constraint C_PSP_FINANCIAL_TRANSACTION2                        check (DEBIT_BANK_ACCOUNT_TYPE IN
('Company', 'Employee', 'Intuit', 'TaxAgency', 'Payee')),
        ON_HOLD NUMBER(1),
        SKU VARCHAR2(40),
        SKU_QUANTITY NUMBER(10),
        BILLING_DETAIL_FK VARCHAR2(255),
        CREDIT_BANK_ACCOUNT_FK VARCHAR2(255),
        DEBIT_BANK_ACCOUNT_FK VARCHAR2(255),
        COMPANY_FK VARCHAR2(255),
        PAYROLL_RUN_FK VARCHAR2(255),
        PAYCHECK_SPLIT_FK VARCHAR2(255),
        TRANSACTION_TYPE_FK VARCHAR2(255) not null
                constraint SYS_C008674                        check ("TRANSACTION_TYPE_FK" IS NOT NULL),
        LAW_FK VARCHAR2(255),
        CURRENT_TRANSACTION_STATE_FK VARCHAR2(255),
        ORIGINAL_TRANSACTION_FK VARCHAR2(255),
        MONEY_MOVEMENT_TRANSACTION_FK VARCHAR2(255),
        ORIGINAL_SETTLEMENT_DATE TIMESTAMP(6),
        REFUND_TYPE VARCHAR2(255 char)
                        constraint C_PSP_FINANCIAL_TRANSACTION3                        check (REFUND_TYPE IN
('ApplyForward', 'Immediate', 'TOR')),
        COMP_ADJUST_SUBMISSION_FK VARCHAR2(255 char),
        TAX_PENALTY_INTEREST_FK VARCHAR2(255 char),
        RELATABLE_TRANSACTION_FK VARCHAR2(255 char),
        BILL_PAYMENT_SPLIT_FK VARCHAR2(255 char),
        COMPANY_LAW_FK VARCHAR2(255 char),
        constraint SYS_C008675
                primary key (FINANCIAL_TRANSACTION_SEQ, REALM_ID),  -- Remove REALM_ID & add COMPANY_FK. NEW PK
(COMAPNY_FK, FINANCIAL_TRANSACTION_SEQ)
    constraint PSP_FINANCIALTRANSACTION_FK1                foreign key (BILLING_DETAIL_FK, REALM_ID) references
PSPADM.PSP_BILLING_DETAIL, -- Remove REALM_ID
    constraint PSP_FINANCIALTRANSACTION_FK2                foreign key (RELATABLE_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION,        -- PK is changing removing REALM_ID and COMPANY_FK to
reflect the same
    constraint PSP_FINANCIALTRANSACTION_FK3                foreign key (ORIGINAL_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION,        -- PK is changing removing REALM_ID and COMPANY_FK to
reflect the same
    constraint PSP_FINANCIAL_TRANSACTION_FK10        foreign key (MONEY_MOVEMENT_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_MONEY_MOVEMENT_TRANSACTION,        -- PK is changing removing REALM_ID and
COMPANY_FK to reflect the same
   constraint PSP_FINANCIAL_TRANSACTION_FK11        foreign key (COMP_ADJUST_SUBMISSION_FK, REALM_ID)
references PSPADM.PSP_COMP_ADJUST_SUBMISSION, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK13        foreign key (BILL_PAYMENT_SPLIT_FK, REALM_ID) references
PSPADM.PSP_BILL_PAYMENT_SPLIT, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK14        foreign key (TAX_PENALTY_INTEREST_FK, REALM_ID) references
PSPADM.PSP_TAX_PENALTY_INTEREST, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK16        foreign key (COMPANY_LAW_FK, REALM_ID) references PSPADM.
PSP_COMPANY_LAW, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK2        foreign key (CREDIT_BANK_ACCOUNT_FK, REALM_ID) references
PSPADM.PSP_BANK_ACCOUNT, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK3        foreign key (DEBIT_BANK_ACCOUNT_FK, REALM_ID) references
```

```
PSPADM.PSP_BANK_ACCOUNT, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK4        foreign key (COMPANY_FK, REALM_ID) references PSPADM.
PSP_COMPANY, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK5        foreign key (PAYROLL_RUN_FK, REALM_ID) references PSPADM.
PSP_PAYROLL_RUN, -- Remove REALM_ID
   constraint PSP_FINANCIAL_TRANSACTION_FK6        foreign key (PAYCHECK_SPLIT_FK, REALM_ID) references PSPADM.
PSP_PAYCHECK_SPLIT,        -- PK is changing removing REALM_ID and COMPANY_FK to reflect the same
    constraint PSP_FINANCIAL_TRANSACTION_FK7        foreign key (TRANSACTION_TYPE_FK, REALM_ID) references
PSPADM.PSP_TRANSACTION_TYPE, -- Remove REALM_ID
    constraint PSP_FINANCIAL_TRANSACTION_FK8        foreign key (LAW_FK, REALM_ID) references PSPADM.PSP_LAW,
-- Remove REALM_ID
    constraint PSP_FINANCIAL_TRANSACTION_FK9        foreign key (CURRENT_TRANSACTION_STATE_FK, REALM_ID)
references PSPADM.PSP_TRANSACTION_STATE -- Remove REALM_ID  )
/
--========================= Child table =========================
--PSP_QBDT_TRANSACTION_INFO
create table PSPADM.PSP_QBDT_TRANSACTION_INFO
(
        QBDT_TRANSACTION_INFO_SEQ VARCHAR2(255 char) not null,
        VERSION NUMBER(19) not null,
        CREATOR_ID VARCHAR2(30 char),
        CREATED_DATE TIMESTAMP(6) not null,
        MODIFIER_ID VARCHAR2(30 char),
        MODIFIED_DATE TIMESTAMP(6) not null,
        REALM_ID NUMBER(19) default -1 not null,
        AGENCY_NAME VARCHAR2(128 char),
        REFERENCE_NUMBER VARCHAR2(11 char),
        ACCOUNT_NAME VARCHAR2(128 char),
        MEMO VARCHAR2(4000 char),
        ON_SERVICE NUMBER(1),
        CLEARED VARCHAR2(1 char),
        TRACKING_CLASS VARCHAR2(128 char),
        IS_DELETED NUMBER(1),
        TOKEN NUMBER(19) default -2,
        LIABILITY_CHECK_FK VARCHAR2(255 char),
        LIABILITY_CHECK_LINE_FK VARCHAR2(255 char),
        MONEY_MOVEMENT_TRANSACTION_FK VARCHAR2(255 char),
        IS_DIRECT_DEPOSIT NUMBER(1) default 0,
        SYSTEM_GENERATED NUMBER(1) default 0,
        COMP_ADJUST_SUBMISSION_FK VARCHAR2(255 char),
        LIABILITY_ADJUSTMENT_FK VARCHAR2(255 char),
        FINANCIAL_TRANSACTION_FK VARCHAR2(255 char),
        PRIOR_PAYMENT_SUBMISSION_FK VARCHAR2(255 char),
        COMPANY_FK VARCHAR2(255 char),
        QBDT_PAYROLL_TRANSACTION_FK VARCHAR2(255 char),
        QBDT_PAYROLL_TRANS_LINE_FK VARCHAR2(255 char),
        primary key (QBDT_TRANSACTION_INFO_SEQ, REALM_ID), -- remove relam_id from PK
        constraint PSP_QBDTTRANSACTIONINFO_FK1                foreign key (PRIOR_PAYMENT_SUBMISSION_FK,
REALM_ID) references PSPADM.PSP_PRIOR_PAYMENT_SUBMISSION,
        constraint PSP_QBDT_TRANSACTION_INFO_FK1                foreign key (LIABILITY_CHECK_FK, REALM_ID)
references PSPADM.PSP_LIABILITY_CHECK,
        constraint PSP_QBDT_TRANSACTION_INFO_FK2                foreign key (LIABILITY_CHECK_LINE_FK, REALM_ID)
references PSPADM.PSP_LIABILITY_CHECK_LINE,
        constraint PSP_QBDT_TRANSACTION_INFO_FK3                foreign key (MONEY_MOVEMENT_TRANSACTION_FK,
REALM_ID) references PSPADM.PSP_MONEY_MOVEMENT_TRANSACTION,-- PK is changing removing REALM_ID and COMPANY_FK
to reflect the same
    constraint PSP_QBDT_TRANSACTION_INFO_FK4                foreign key (FINANCIAL_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION,-- PK is changing removing REALM_ID and COMPANY_FK to reflect the
same
        constraint PSP_QBDT_TRANSACTION_INFO_FK5                foreign key (COMP_ADJUST_SUBMISSION_FK,
REALM_ID) references PSPADM.PSP_COMP_ADJUST_SUBMISSION,
        constraint PSP_QBDT_TRANSACTION_INFO_FK6                foreign key (LIABILITY_ADJUSTMENT_FK, REALM_ID)
references PSPADM.PSP_LIABILITY_ADJUSTMENT,
        constraint PSP_QBDT_TRANSACTION_INFO_FK7                foreign key (COMPANY_FK, REALM_ID) references
PSPADM.PSP_COMPANY,
        constraint PSP_QBDT_TRANSACTION_INFO_FK8                foreign key (QBDT_PAYROLL_TRANSACTION_FK,
REALM_ID) references PSPADM.PSP_QBDT_PAYROLL_TRANSACTION,
        constraint PSP_QBDT_TRANSACTION_INFO_FK9                foreign key (QBDT_PAYROLL_TRANS_LINE_FK,
REALM_ID) references PSPADM.PSP_QBDT_PAYROLL_TRANS_LINE
)
/
```

```
create index PSPADM.PSP_QBDTTRANSACTIONINFO_FK1 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(PRIOR_PAYMENT_SUBMISSION_FK, REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK1 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_CHECK_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK2 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_CHECK_LINE_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK3 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(MONEY_MOVEMENT_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg (COMPANY_FK,
MONEY_MOVEMENT_TRANSACTION_FK)
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK4 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(FINANCIAL_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg (COMPANY_FK,
FINANCIAL_TRANSACTION_FK)
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK5 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(COMP_ADJUST_SUBMISSION_FK, REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK6 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_ADJUSTMENT_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK7 on PSPADM.PSP_QBDT_TRANSACTION_INFO (COMPANY_FK, REALM_ID); --
remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK8 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(QBDT_PAYROLL_TRANSACTION_FK, REALM_ID); --remove REALM_ID
create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK9 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(QBDT_PAYROLL_TRANS_LINE_FK, REALM_ID); --remove REALM_ID

--PSP_TRANSACTION_OFFLOAD_BATCH
create table PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH
(
        TRANSACTION_OFFLOAD_BATCH_SEQ VARCHAR2(255 char) not null,
        VERSION NUMBER(19) not null,
        CREATOR_ID VARCHAR2(30 char),
        CREATED_DATE TIMESTAMP(6) not null,
        MODIFIER_ID VARCHAR2(30 char),
        MODIFIED_DATE TIMESTAMP(6) not null,
        REALM_ID NUMBER(19) default -1 not null,
        OFFLOAD_STATUS_CD VARCHAR2(10 char),
        STATUS_EFFECTIVE_DATE TIMESTAMP(6),
        FINANCIAL_TRANSACTION_FK VARCHAR2(255 char) not null,
        OFFLOAD_BATCH_FK VARCHAR2(255 char) not null,
        primary key (TRANSACTION_OFFLOAD_BATCH_SEQ, REALM_ID), --Remove REALM_ID
constraint PSP_TRANSACTION_OFFLOAD_BA_FK1                foreign key (FINANCIAL_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_FINANCIAL_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the
same. Eg (COMPANY_FK,FINANCIAL_TRANSACTION_FK)
  constraint PSP_TRANSACTION_OFFLOAD_BA_FK2                foreign key (OFFLOAD_BATCH_FK, REALM_ID) references
PSPADM.PSP_OFFLOAD_BATCH --remove REALM_ID
)
/

create index PSPADM.PSP_TRANSACTION_OFFLOAD_BA_FK2 on PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH (OFFLOAD_BATCH_FK,
REALM_ID); --remove REALM_ID
create index PSPADM.PSP_TRANSACTION_OFFLOAD_BA_FK1 on PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH
(FINANCIAL_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg (COMPANY_FK,
FINANCIAL_TRANSACTION_FK)
```

Partitioning Principles

- Consider table for partitioning whose size is >100G. This is to handle organic & inorganic growth for tables.
- Partition tables by tenant id (company_fk). This will also help in horizontal scalability of database.
- Consider tables sizes and growth rate (estimated/projected next 3 year and 5 year)
- Keep same number of partitions for logical grouping of tables as per workflows
- Keep individual partition size less than or equal to 50G
- Distribute tables across multiple tablespaces to ensure we do not hit AWS RDS big data file size limit.
- Review partitioned tables for OLTP & OLAP traffic to confirm on number of partitions
- partitioned tables should have company_fk filter

Query Analysis POC

- Partitioned Table Join With Non Partitioned Table

Filter /Join	Queries	Query Plan	Observations
Both + company_fk join	explain select * from pspadmhashdms. psp_money_movement_transaction mmt, pspadmhashdms. psp_atfpayments_to_process atp where mmt. money_movement_transaction_seq = atp. money_movement_transaction_fk and mmt.COMPANY_FK ='81ea198c-6fa4-4a19-abcd-705c93605e43' and atp.COMPANY_FK ='81ea198c-6fa4-4a19-abcd-705c93605e43' ; explain select * from pspadmhashdms. psp_money_movement_transaction mmt, pspadmhashdms. psp_atfpayments_to_process atp where mmt. money_movement_transaction_seq = atp. money_movement_transaction_fk and mmt.company_fk = atp. company_fk and mmt.COMPANY_FK ='81ea198c-6fa4-4a19-abcd-705c93605e43' and atp.COMPANY_FK ='81ea198c-6fa4-4a19-abcd-705c93605e43' ;	QUERY PLAN -- Merge Join (cost=1.39..3967.18 rows=2 width=1352) Merge Cond: ((mmt.money_movement_transaction_seq)::text = (atp.money_movement_transaction_fk)::text) -> Index Scan using psp_money_movement_transaction_hash_p3_pkey on psp_money_movement_transaction_p3 mmt (cost=0.70..2589.48 rows=2502 width=1153) Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19-abcd-705c93605e43'::text) -> Index Scan using psp_atfpayments_to_process_fk1 on psp_atfpayments_to_process atp (cost=0.69..1368.09 rows=1337 width=199) Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19-abcd-705c93605e43'::text) (6 rows)	Company_fk join is not required.

- Partitioned Table Join With Partitioned Table

Filter /Join	Queries	Query Plan	Observations

Both + company_fk join	select * from pspadmhashdms.psp_money_movement_transaction mmt, pspadmhashdms.psp_financial_transaction ft where mmt.money_movement_transaction_seq = ft.money_movement_transaction_fk and mmt.COMPANY_FK = ft.COMPANY_FK and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43' and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43'; select * from pspadmhashdms.psp_money_movement_transaction mmt, pspadmhashdms.psp_financial_transaction ft where mmt.money_movement_transaction_seq = ft.money_movement_transaction_fk and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43' and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43';	same plan for both queries QUERY PLAN -- Merge Join (cost=1.39..4806.82 rows=22756 width=3289) Merge Cond: ((ft.money_movement_transaction_fk)::text = (mmt.money_movement_transaction_seq)::text) -> Index Scan using psp_financial_transaction_fk10_p3 on psp_financial_transaction_p3 ft (cost=0.69..1871.46 rows=1819 width=2136) Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19-abcd-705c93605e43'::text) -> Materialize (cost=0.70..2595.73 rows=2502 width=1153) -> Index Scan using psp_money_movement_transaction_hash_p3_pkey on psp_money_movement_transaction_p3 mmt (cost=0.70..2589.48 rows=2502 width=1153) Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19-abcd-705c93605e43'::text) (7 rows)	Company_fk join is not required.

- 2 Partitioned Tables Join With Non-Partitioned Table

2 Partitioned /Non-partition tables filters /join	explain select *
	from pspadmhashdms.psp_money_movement_transaction mmt,
	     pspadmhashdms.psp_financial_transaction ft,
	     pspadmhashdms.psp_atfpayments_to_process atp
	where mmt.money_movement_transaction_seq = ft.money_movement_transaction_fk
	  and mmt.money_movement_transaction_seq = atp.money_movement_transaction_fk
	  and mmt.COMPANY_FK = ft.COMPANY_FK
	  and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43'
	  and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43'
	  and atp.COMPANY_FK ='81ea198c-6fa4-4a19-abcd-705c93605e43';
	
	explain select *
	from pspadmhashdms.psp_money_movement_transaction mmt,
	     pspadmhashdms.psp_financial_transaction ft,
	     pspadmhashdms.psp_atfpayments_to_process atp
	where mmt.money_movement_transaction_seq = ft.money_movement_transaction_fk
	  and mmt.money_movement_transaction_seq = atp.money_movement_transaction_fk
	  and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43'
	  and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd-705c93605e43'
	  and atp.COMPANY_FK ='81ea198c-6fa4-4a19-abcd-705c93605e43';
		QUERY PLAN
		--- 	
		 Nested Loop (cost=2.08..3250.17 rows=18 width=3488)
		   -> Merge Join (cost=1.38..3247.45 rows=1 width=2335)
		         Merge Cond: ((ft.money_movement_transaction_fk)::text = (atp.money_movement_transaction_fk)::text)
		         -> Index Scan using psp_financial_transaction_fk10_p3 on psp_financial_transaction_p3 ft (cost=0.69..1871.46 rows=1819 width=2136)
		               Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19-abcd-705c93605e43'::text)
		         -> Index Scan using psp_atfpayments_to_process_fk1 on psp_atfpayments_to_process atp (cost=0.69..1368.09 rows=1337 width=199)
		               Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19-abcd-705c93605e43'::text)
		   -> Index Scan using psp_money_movement_transaction_hash_p3_pkey on psp_money_movement_transaction_p3 mmt (cost=0.70..2.71 rows=1 width=1153)
		         Index Cond: (((company_fk)::text = '81ea198c-6fa4-4a19-abcd-705c93605e43'::text) AND ((money_movement_transaction_seq)::text = (ft.money_movement_transaction_fk)::text))
		(9 rows)
			Company _fk join is not required.

Schema Conversion

Data Type Mapping

Criteria followed for target datatype w.r.t source NUMER datatype

Precision(m)	Scale(n)	Oracle	PostgreSQL	Comments
<5	0	NUMBER(m,n)	SMALLINT	Documentation Page
5 >= m <= 9	0	NUMBER(m,n)	INT	
10<= m <=18	0	NUMBER(m,n)	BIGINT	
m+n <= 15	n>0	NUMBER(m,n)	DOUBLE PRECISION	
m+n > 15	n>0	NUMBER(m,n)	NUMERIC	

NUMBER(19,4) and NUMBER(19,7) Decision

- Use NUMERIC with same precision and scale as that of Oracle

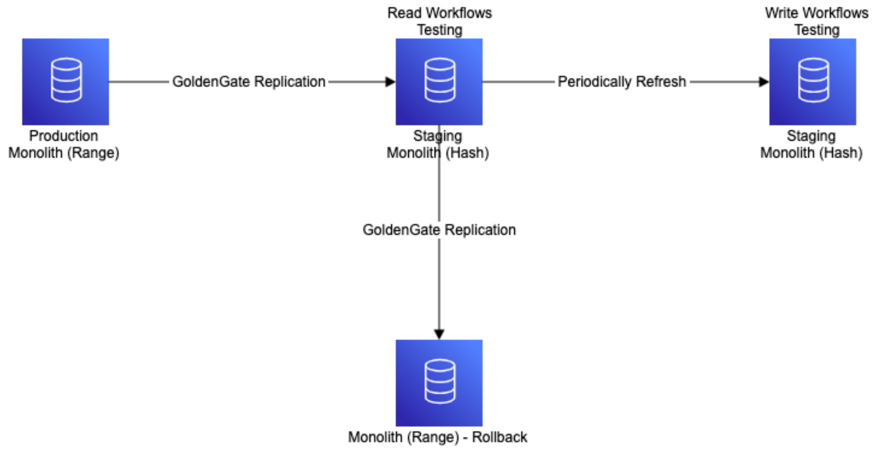
NUMBER(10,0) and NUMBER(19,0) Optimization

There is opportunity to optimize numeric(10) and numeric(19) data types further. Based on actual values (Min, Max) data analysis, we are following below criteria for these two

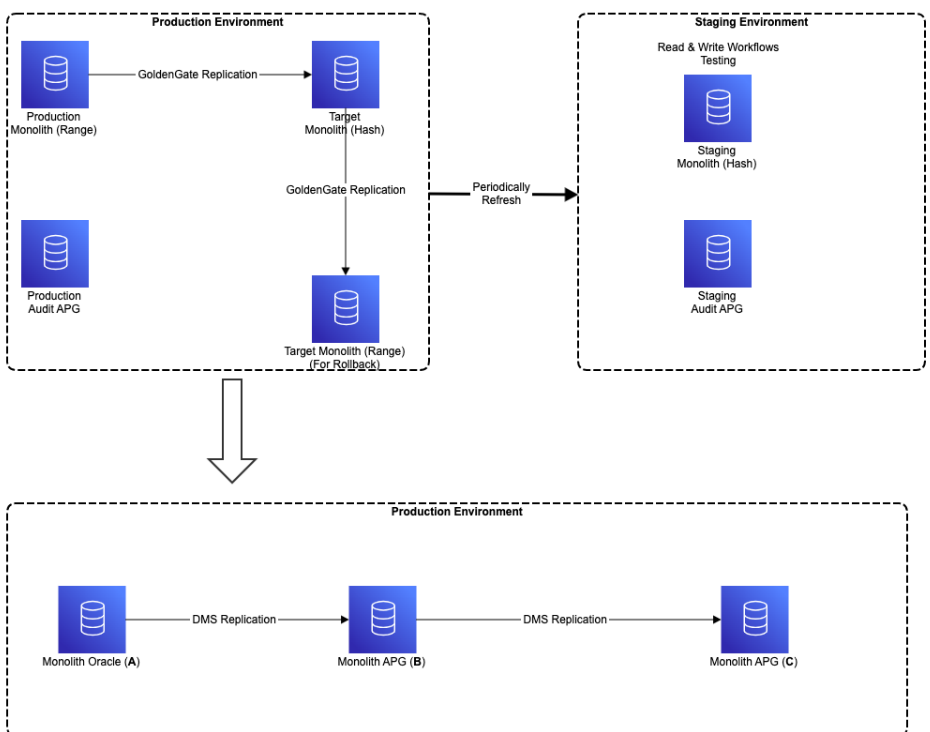
- if source datatype is number(10) and current Max value is more than 100M then use BIGINT else use INTEGER
- if source datatype is number(19) and current Max value is more than 100M. Using NUMERIC(19) else use INTEGER
- Use SMALLINT data type in Postgres for NUMBER(1) in Oracle.
- Use INTEGER data type in Postgres for all VERSION columns in all tables to maintain consistency. This has NUMBER(19) data type in Oracle.
- Using SMALLINT for REALM_ID column in all tables. We used for Audit db migration as well.

	Oracle	Postgres
1	CLOB	TEXT
2	NUMBER(1)	SMALLINT
3	NUMBER(10)	BIGINT
4	NUMBER(19)	NUMERIC(19,0)
5	NUMBER(19) NOT NULL	NUMERIC(19,0) NOT NULL
6	NUMBER(19,7)	NUMERIC(19,7)
7	TIMESTAMP(6)	timestamp(6) without time zone
8	TIMESTAMP(6) NOT NULL	timestamp(6) without time zone NOT NULL
9	VARCHAR2(100 Char)	character varying(100)
10	VARCHAR2(20 Char)	character varying(20)
11	VARCHAR2(200 Char)	character varying(200)
12	VARCHAR2(255 Char)	character varying(255)
13	VARCHAR2(255 Char) NOT NULL	character varying(255) NOT NULL
14	VARCHAR2(256 Char)	character varying(256)
15	VARCHAR2(30 Char)	character varying(30)
16	VARCHAR2(40 Char)	character varying(40)
17	VARCHAR2(400 Char)	character varying(400)
18	VARCHAR2(4000 Char)	character varying(4000)

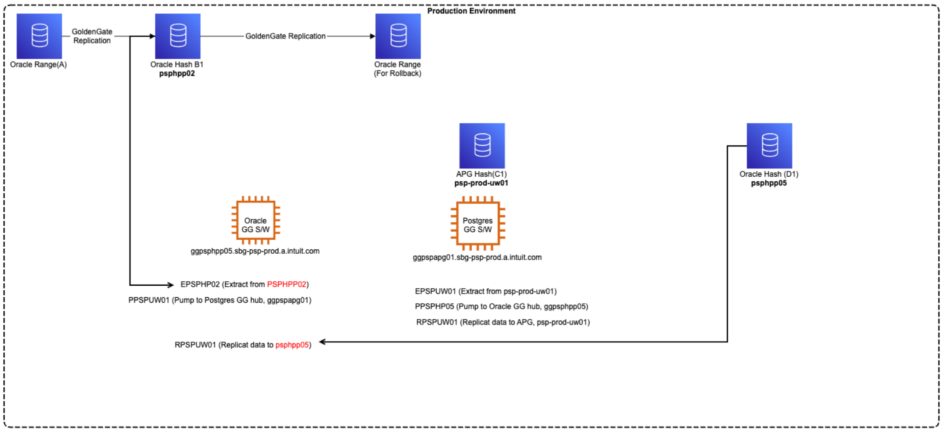
Production Replication and Staging Refresh Strategy



Read Workflows Testing
Write Workflows Testing
GoldenGate Replication
Periodically Refresh
Production Monolith (Range)
Staging Monolith (Hash)
Staging Monolith (Hash)
GoldenGate Replication
Monolith (Range) - Rollback



Production Environment
GoldenGate Replication
Production Monolith (Range)
Target Monolith (Hash)
GoldenGate Replication
Production Audit APG
Target Monolith (Range) (For Rollback)
Periodically Refresh
Staging Environment
Read & Write Workflows Testing
Staging Monolith (Hash)
Staging Audit APG
Production Environment
DMS Replication
DMS Replication
Monolith Oracle (A)
Monolith APG (B)
Monolith APG (C)


Production Environment
Oracle Range(A)
GoldenGate Replication
Oracle Hash B1
psphpp02
GoldenGate Replication
Oracle Range
(For Rollback)
APG Hash(C1)
psp-prod-uw01
Oracle Hash (D1)
psphpp05
Oracle
GG S/W
ggpsphpp05.sbg-psp-prod.a.intuit.com
Postgres
GG S/W
ggpspapg01.sbg-psp-prod.a.intuit.com
EPSPHP02 (Extract from PSPHPP02)
PPSPUW01 (Pump to Postgres GG hub, ggpspapg01)
EPSPUW01 (Extract from psp-prod-uw01)
PPSPHP05 (Pump to Oracle GG hub, ggpsphpp05)
RPSPUW01 (Replicat data to APG, psp-prod-uw01)
RPSPUW01 (Replicat data to psphpp05)